GAZİ UNIVERSITY

DEPARTMENT OF COMPUTER ENGINEERING

COMPUTER GRAPHICS

Project Proposal

Project Name: Pitstop Challenge

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1. Brief Introduction

This project presents an interactive 3D Formula 1 pit garage simulation developed strictly using WebGL. An F1 car enters the pit lane and the user, navigating freely through the garage environment, performs component replacements by picking up parts and attaching them to the vehicle. The project demonstrates hierarchical modeling, proximity-based snapping, dynamic lighting, and texture mapping entirely through low-level WebGL and GLSL, without relying on any game engine or graphics framework.

2. Detailed Description

The scene is set inside a Formula 1 pit garage. When the simulation begins, an F1 car rolls into the pit lane and stops. The user navigates the 3D environment using a free-roaming first-person camera (WASD keys and mouse look), able to move freely along all three axes and rotate in any direction. The garage contains loose components — tires and a front wing — placed on the floor that the user can pick up and carry.

The scene contains at least three distinct object types with different morphologies: the F1 car body (a complex multi-part mesh), cylindrical tires, and the rectangular garage environment (walls, floor, toolboxes). Each of these carries a different texture: a carbon-fiber/metallic paint texture on the car body, a matte rubber texture on the tires, and a concrete/asphalt texture on the garage floor, satisfying both the morphology and texture requirements.

The three independently controllable objects in the scene are the tire, the front wing, and the rear wing flap, each serving a distinct interaction:

Pit Stop Component Replacement: The user picks up a tire or front wing from the garage floor using a key press and carries it toward the car. When the held component is brought sufficiently close to its mounting point on the vehicle, it automatically snaps into place, updating the car's model state. This proximity-based snapping is implemented via distance checks in the transformation matrix pipeline.

Cockpit View & Steering Synchronization: The user can move the camera inside the driver's cockpit. From this viewpoint, pressing the left/right keys rotates the steering wheel. Using a parent-child hierarchy in the scene's transformation matrix stack, the front tires are linked as child nodes of the steering wheel, so their yaw rotation stays synchronized with the wheel's rotation automatically.

Drag Reduction System (DRS): A dedicated key press toggles the DRS. The rear wing flap, modeled as a separate mesh segment, rotates along its local X-axis to its open or closed position, demonstrating independent local transformations on a child node within the car's hierarchical model.

Lighting is handled via the Phong reflection model implemented in GLSL fragment shaders. A main garage spotlight serves as the primary light source; the user can reposition it across the scene using keyboard controls and adjust its brightness up or down, producing visible specular highlights and shading changes on the car's metallic surfaces in real time.